

DBMS SELLER AND INVENTORY MANAGEMENT

ZEPTO E-COMMERCE

1. Database Design:

Seller Table

Attribute	Description
SELLER_ID	Unique seller ID
SELLER_NAME	Seller name
EMAIL	Seller email address
PHONE_NUMBER	Seller contact number
ADDRESS	Seller address

Inventory Table

Attribute	Description
INVENTORY_ID	Unique inventory ID
SELLER_ID	Seller ID
PRODUCT_ID	Product ID
STOCK_QUANTITY	Available stock quantity
STOCK_STATUS	Product availability status

2. Define Primary and Foreign Key Relationships:

Seller → Inventory

- SELLER.SELLER_ID → Primary Key
- INVENTORY.INVENTORY_ID → Primary Key
- INVENTORY.SELLER_ID → Foreign Key referencing SELLER.SELLER_ID
- INVENTORY.PRODUCT_ID → Foreign Key referencing PRODUCT.PRODUCT_ID

- Relationship: One-to-Many (1:M)

One seller can manage many inventory records.

The Inventory table connects Seller and Product and maintains stock quantity and availability status.

3. Table Creation:

Create Seller Table:

```
SQL> CREATE TABLE SELLER(  
 2     SELLER_ID NUMBER PRIMARY KEY,  
 3     SELLER_NAME VARCHAR2(30),  
 4     EMAIL VARCHAR2(30),  
 5     PHONE_NUMBER VARCHAR2(15),  
 6     ADDRESS VARCHAR2(80)  
 7 );
```

Table created.

Create Inventory Table:

```
SQL> CREATE TABLE INVENTORY(  
 2     INVENTORY_ID NUMBER PRIMARY KEY,  
 3     SELLER_ID NUMBER,  
 4     PRODUCT_ID NUMBER,  
 5     STOCK_QUANTITY NUMBER,  
 6     STOCK_STATUS VARCHAR2(20),  
 7     FOREIGN KEY(SELLER_ID) REFERENCES SELLER(SELLER_ID),  
 8     FOREIGN KEY(PRODUCT_ID) REFERENCES PRODUCT(PRODUCT_ID)  
 9 );
```

Table created.

4. Data Insertion

Seller Records

```
SQL> INSERT INTO SELLER VALUES(201,'FRESH MART','freshmart@gmail.com','9876543210','Anna Nagar, Chennai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES(202,'DAILY NEEDS','dailyneeds@gmail.com','8765432109','Tambaram, Chennai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES(203,'GREEN STORE','greenstore@gmail.com','7986543210','Velachery, Chennai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES(204,'CITY MART','citymart@gmail.com','7654321098','Adyar, Chennai');
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO SELLER VALUES(205,'SMART GROCERY','smartgrocery@gmail.com','9123456780','T Nagar, Chennai');
```

1 row created.

Inventory Records

```
SQL> INSERT INTO INVENTORY VALUES(301,201,101,50,'AVAILABLE');

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES(302,202,102,80,'AVAILABLE');

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES(303,203,103,70,'AVAILABLE');

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES(304,204,104,90,'AVAILABLE');

1 row created.

SQL>
SQL> INSERT INTO INVENTORY VALUES(305,205,105,0,'UNAVAILABLE');

1 row created.
```

5.Data Retrieval:

Display Seller Records

```
SQL> SELECT * FROM SELLER;

SELLER_ID SELLER_NAME          EMAIL
-----
PHONE_NUMBER
-----
ADDRESS
-----
| | | 201 FRESH MART          freshmart@gmail.com
9876543210
Anna Nagar, Chennai
| | | 202 DAILY NEEDS        dailyneeds@gmail.com
8765432109
Tambaram, Chennai
| | | 203 GREEN STORE        greenstore@gmail.com
7986543210
Velachery, Chennai
| | | 204 CITY MART          citymart@gmail.com
7654321098
| | | 205 SMART GROCERY      smartgrocery@gmail.com
9123456780
T Nagar, Chennai
```

Display Inventory Records

```
SQL> SELECT * FROM INVENTORY;
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
301	201	101	50	AVAILABLE
302	202	102	80	AVAILABLE
303	203	103	70	AVAILABLE
304	204	104	90	AVAILABLE
305	205	105	0	UNAVAILABLE

```
SQL>
```

Display Seller Product Information

```
SQL> SELECT S.SELLER_NAME,  
2         P.PRODUCT_NAME,  
3         I.STOCK_QUANTITY,  
4         I.STOCK_STATUS  
5 FROM SELLER S  
6 JOIN INVENTORY I  
7 ON S.SELLER_ID = I.SELLER_ID  
8 JOIN PRODUCT P  
9 ON I.PRODUCT_ID = P.PRODUCT_ID;
```

SELLER_NAME	PRODUCT_NAME	STOCK_QUANTITY
FRESH MART AVAILABLE	APPLE	50

DAILY NEEDS AVAILABLE	BANANA	80
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GREEN STORE AVAILABLE	TOMATO	70
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SELLER_NAME	PRODUCT_NAME	STOCK_QUANTITY
CITY MART AVAILABLE	MILK	90

SMART GROCERY UNAVAILABLE	BISCUIT	0
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6. Track Available Products

```
SQL> SELECT P.PRODUCT_NAME,  
2      I.STOCK_QUANTITY,  
3      I.STOCK_STATUS  
4 FROM PRODUCT P  
5 JOIN INVENTORY I  
6 ON P.PRODUCT_ID = I.PRODUCT_ID  
7 WHERE I.STOCK_STATUS = 'AVAILABLE';
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
APPLE	50	AVAILABLE
BANANA	80	AVAILABLE
TOMATO	70	AVAILABLE
MILK	90	AVAILABLE

7. Track Unavailable Products

```
SQL> SELECT P.PRODUCT_NAME,  
2      I.STOCK_QUANTITY,  
3      I.STOCK_STATUS  
4 FROM PRODUCT P  
5 JOIN INVENTORY I  
6 ON P.PRODUCT_ID = I.PRODUCT_ID  
7 WHERE I.STOCK_STATUS = 'UNAVAILABLE';
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
BISCUIT	0	UNAVAILABLE

8. Inventory Status Report

```
SQL> SELECT S.SELLER_NAME,  
2      P.PRODUCT_NAME,  
3      I.STOCK_QUANTITY,  
4      I.STOCK_STATUS  
5 FROM SELLER S  
6 JOIN INVENTORY I  
7 ON S.SELLER_ID = I.SELLER_ID  
8 JOIN PRODUCT P  
9 ON I.PRODUCT_ID = P.PRODUCT_ID  
10 ORDER BY I.STOCK_STATUS, S.SELLER_NAME;
```

SELLER_NAME	PRODUCT_NAME	STOCK_QUANTITY
CITY MART AVAILABLE	MILK	90
DAILY NEEDS AVAILABLE	BANANA	80
FRESH MART AVAILABLE	APPLE	50

SELLER_NAME	PRODUCT_NAME	STOCK_QUANTITY
GREEN STORE AVAILABLE	TOMATO	70
SMART GROCERY UNAVAILABLE	BISCUIT	0